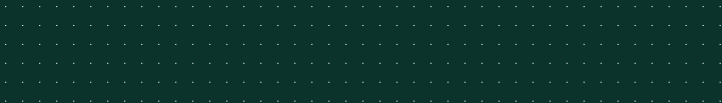


JUNE 25, 2026

Leveraging Local LLMs for Automated Changelog Generation

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Outline

Introduction

T5 Model

Data Collection Pipeline

Fine-Tuning & Results

Quantization & Deployment



Quickstart

Tumbleweed

```
1 #> sudo zypper in python-changelog-gen
2 #> cd SUPER_DUPER_PKG
3 #> osc-ai-vc
```

Leap 16.0

```
1 #> sudo zypper ar obs://science:machinelearning sc:ml
2 #> sudo zypper in python-changelog-gen
3 #> cd SUPER_DUPER_PKG
4 #> osc-ai-vc
```

Resources

<https://huggingface.co/mslacken/t5-finetune-changelog>

<https://github.com/openSUSE/changelog-gen>



Introduction

Motivation

Changelogs in Open Build Service

- Package changelogs are created from `.changes` files
- Mostly reflect upstream changelog
- **But** must also contain changes for package creation

The Opportunity

- Accepted requests to `openSUSE:Factory` **are** the right format
- Submission requests contain the necessary information
- Perfect candidate for LLM automation



Why use a Local LLM?

Existing Tools

- Existing tools can do the same
- Cannot capture changes to `.spec` or `_multibuild` files

Advantages

- **Hype Train**
- Pick up semantic changes (e.g., avoid Apple-specific changes)
- Runs completely locally, handling embargoed changes
- No information leak to cloud models



Which Model to Choose?

Why Fine-Tune Instead of Training from Scratch?

- Pre-trained models understand language structure
- Training from scratch requires massive datasets (billions of tokens)
- Fine-tuning is computationally cheaper and faster
- Specializes existing knowledge for our specific task

Constraints

- Model must be small (around 1 GB)
- Permissive license
- Not quantized, so that it can be fine-tuned
- Transformer-based and not a GPT model



T5 Model

Encoder-Decoder vs. Decoder-Only

Encoder-Decoder: T5, BART, mT5

- **Bidirectional encoder:** Sees entire input at once
- Best for: Translation, summarization, structured text generation

Decoder-Only: GPT-3, GPT-4, LLaMA, Mistral

- **Single decoder:** Input and output in same stream, but only sees previous tokens
- Best for: Text completion, general-purpose chat

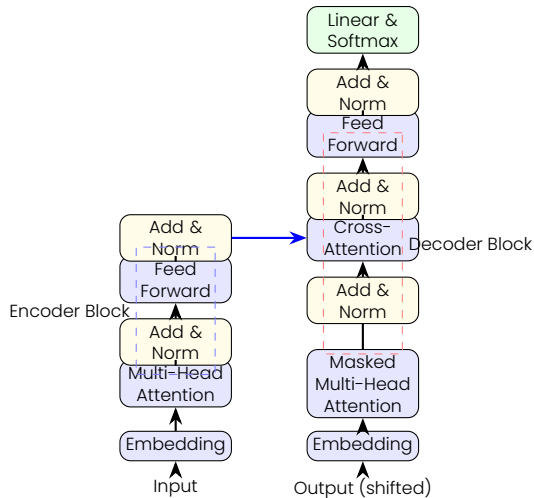
Key Difference

T5 can **read all input first**, then generate. GPT processes input and output sequentially. This makes T5 more efficient for tasks with structured input.



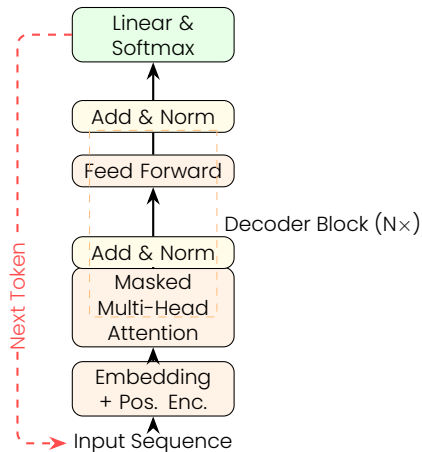
Encoder-Decoder Architecture (T5)

- **Encoder:** Bidirectional processing of the input sequence
- **Decoder:** Autoregressive generation with cross-attention to encoder



Decoder-Only Architecture (GPT/LLaMA)

- **Streamlined:** Unified architecture for both input and output
- **Causal Masking:** Each token only attends to previous tokens



Attention Mechanisms: Full vs. Masked

Full Attention (T5 Encoder)

create	0.21	0.12	0.13	0.08	0.06	0.15	0.05	0.2
changelog	0.16	0.17	0.14	0.15	0.05	0.07	0.13	0.13
for	0.14	0.19	0.05	0.09	0.18	0.11	0.12	0.11
pkg	0.15	0.16	0.06	0.19	0.09	0.11	0.1	0.13
gnome	0.08	0.13	0.12	0.06	0.2	0.11	0.16	0.14
music	0.22	0.08	0.07	0.17	0.1	0.14	0.07	0.15
44	0.17	0.17	0.11	0.04	0.18	0.13	0.04	0.16
45	0.08	0.18	0.18	0.07	0.08	0.18	0.1	0.13
	create	changelog	for	pkg	gnome	music	44	45

All tokens considered

Masked Attention (GPT)

create	1							
changelog	0.29	0.71						
for	0.28	0.3	0.41					
pkg	0.25	0.2	0.23	0.32				
gnome	0.21	0.12	0.17	0.2	0.29			
music	0.19	0.18	0.24	0.13	0.1	0.16		
44	0.1	0.09	0.25	0.18	0.08	0.08	0.23	
45	0.16	0.19	0.1	0.14	0.05	0.11	0.18	0.07
	create	changelog	for	pkg	gnome	music	44	45

Only previous tokens considered

Numbers show attention weights (sum to 1.0 per row)

Darker green = stronger attention.



T5: Origins and History

The T5 Paper (2019) by Google

- **T5 = Text-To-Text Transfer Transformer**
- Systematic study of NLP transfer learning
- **Everything is text-to-text**
- Apache 2.0 License

Pre-training

- Trained on C4 dataset (Colossal Clean Crawled Corpus)
- 750 GB of cleaned web text



T5 Architecture: Encoder-Decoder

Encoder-Decoder Design

- **Encoder:** Processes input text
 - Bidirectional attention
 - Understands full context
 - Creates rich representations
- **Decoder:** Generates output text
 - Autoregressive generation
 - Attends to encoder outputs
 - Produces token-by-token

Model Sizes

Model	Params	Layers	Size
Small	60M	12	242 MB
Base	220M	24	892 MB
Large	770M	48	2.75 GB



Data Collection Pipeline

OBS Data Extraction

What data do we need?

- Accepted requests to `openSUSE:Factory`
- Spec file diffs (version updates, build changes)
- Source file changes (added, removed, modified)
- Release notes from upstream (GitHub, archive changelogs)
- The actual `.changes` file entries (our target)

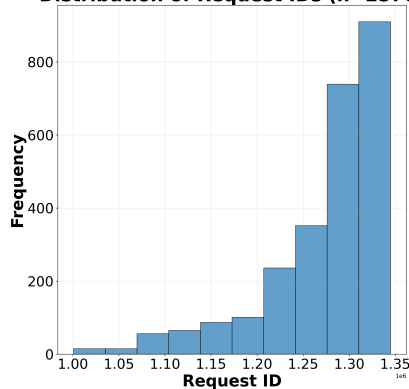


OBS Data Extraction

Naive Approach

- Choose a random number smaller than the latest request
- Check if submission went to `openSUSE:Factory` and was accepted
- Identify `Source0`
- Extract first 20 lines of `CHANGELOG` ...
- For GitHub links, fetch GitHub release page

Distribution of Request IDs (n=2576)



Use half-side Gaussian to get recent requests



OBS Data Extraction

Privacy Considerations

- Spec files use the MIT license; content can be freely used
- Changes files also use the MIT license
- CHANGELOG files fall under OSS licenses if accepted to openSUSE :Factory
- GitHub release pages use the same license as the source

Personal Information

- Submitter name and email are removed from changes files
- Reviewer information is not retrieved
- Request IDs are used as keys but not for training



OBS Data Extraction – Example Update

```
1 { "request_id": 1111873, "project": "GNOME:Factory", "package": "gnome-music", "version": "45.0",
2   "changed_files": [ "gnome-music-45.0.tar.xz", "gnome-music.changes", "gnome-music.spec" ],
3   "removed_files": [],
4   "added_files": [],
5   "spec_diff": "Name:          gnome-music\n-Version:          44.0\n+Version:          45.0\n Release:
   0\n Summary:          Music Player for GNOME\n License:          LGPL-2.1-or-later AND SUSE-GPL-2.0-with-
   plugin-exception\n Group:          Productivity/Multimedia/Sound/Players\n URL:          https://www.
   gnome.org\n-Source0:          https://download.gnome.org/sources/gnome-music/44/{name}-{version}.tar.xz\n
   +Source0:          https://download.gnome.org/sources/gnome-music/45/{name}-{version}.tar.xz\n #PATCH-FIX
   -SLE 0001-gnome-music-use-python36.patch yfjiang@suse.com -- disable python 3.7 specific feature to
   allow gnome-music build and run on python 3.6 for SLE/Leap 15.4.\n Patch0:          0001-gnome-music-use-
   python36.patch",
6   "changes_diff": "+\n+- Update to version 45.0:\n+  + Minor quality of life improvements.\n+  + Updated
   translations.\n+\n- Update to version 44.0:",
7   "source_file": "gnome-music-45.0.tar.xz",
8   "archive_changelog": "Overview of changes in 45.0\n=====
   \n\n Minor quality of life
   improvements\n\nThanks to our contributors this release:\n Carlos Garnacho\n Jean Felder\n Harold
   Knowlden\n Marinus Schraal\nUpdated or added translations:\n Abkhazian\n Romanian\n Italian\n Turkish\n
   British English\n French\n Hungarian\n German\n Hebrew\n Ukrainian",
9   "url": "https://www.gnome.org",
10  "github_release_notes": "" },
```



OBS Data Extraction – Example Update Rendered

```
1 <INPUT>
2 create structured changelog for package gnome-music from 44.0 to 45.0:
3 changelog:
4 Overview of changes in 45.0
5 =====
6 • Minor quality of life improvements
7 Thanks to our contributors this release:
8 Carlos Garnacho
9 Jean Felder
10 Harold Knowlden
11 Marinus Schraal
12 Updated or added translations:
13 Abkhazian
14 [...]
15 Hebrew
16 Ukrainian
17 changes in spec file:
18 -Source0: https://download.gnome.org/sources/gnome-music/44/%<unk>name<unk>-%<unk>version<unk>.tar.xz
19 +Source0: https://download.gnome.org/sources/gnome-music/45/%<unk>name<unk>-%<unk>version<unk>.tar.xz
20 </INPUT>
21 -----
22 <TARGET>
23 - Update to version 45.0:
24   + Minor quality of life improvements.
25   + Updated translations.
26 </TARGET>
```



OBS Data Extraction – Example Patch File

```
1 { "request_id": 1071207, "project": "network:utilities", "package": "pdsh", "version": "2.34",
2   "changed_files": [ "pdsh.changes", "pdsh.spec" ],
3   "removed_files": [],
4   "added_files": [ "Add-call-to-slurm_init-this-makes-sure-the-config-options-are-set.patch" ],
5   "spec_diff": "Patch6:          fail-fast-on-ssh-errors-or-non-zero-return-code.patch\n Patch7:          doc-
fast-fail-update.patch\n Patch8:          Hack-to-work-around-a-generic-type-name-breakage-introduced-by-
latest-Slurm.patch\n+Patch9:          Add-call-to-slurm_init-this-makes-sure-the-config-options-are-set.
patch\n %description\n Pdsh is a multithreaded remote shell client which executes commands on\n %patch6 -
p1\n %patch7 -p1\n %patch8 -p1\n+ %patch9 -p1\n %build\n export CFLAGS=\"%{optflags} -fno-strict-aliasing
\"",
6   "changes_diff": "+\n+- Fix slurm plugin: make sure slurm_init() is called before using\n+ the Slurm API.
This has been the case since version 20.11 (bsc#1209216).\n+ Add-call-to-slurm_init-this-makes-sure-the-
config-options-are-set.patch\n+\n - Hack-to-work-around-a-generic-type-name-breakage-introduced-by-latest
-Slurm.patch",
7   "source_file": "pdsh-2.34.tar.gz",
8   "archive_changelog": "This file describes changes in recent versions of pdsh. It primarily\ndocuments
those changes that are of interest to users and admins.\n\n* Changes in pdsh-2.34 (2020-01-07)\n
=====\n -- Fix for output corruption with no newlines (#114)\n -- pipecmd:
fix result check error handling (Dylan Simon)\n -- slurm: workaround slurm export of internal List
interfaces\n -- readline: add application name as argv[0] (#112)\n -- Fix errors from lgtm.com scan\n\n*
Changes in pdsh-2.33 (2017-06-28)\n===== \n -- Fix segfault and build issues
on Mac OSX (#95)\n -- Always pass RTLD_GLOBAL to dlopen(3) of modules. Fixes missing symbol\n errors
from modules using libraries that also use dlopen() (e.g.\n nodeupdown, slurm)\n\n* Changes in pdsh-
2.32 (2017-06-22)\n===== \n -- Autotools update\n -- Switch to dlopen(3)/
dlsym(3) instead of using libltdl",
9   "url": "http://pdsh.googlecode.com/",
10  "github_release_notes": ""
11
```



OBS Data Extraction – Example Patch File Rendered

```
1 <INPUT>
2 create structured changelog for package pdsh 2.34:
3 new files: ['Add-call-to-slurm_init-this-makes-sure-the-config-options-are-set.patch']
4 changes in spec file:
5 +Patch9: Add-call-to-slurm_init-this-makes-sure-the-config-options-are-set.patch
6 +%patch9 -p1
7 </INPUT>
8 -----
9 <TARGET>
10 - Fix slurm plugin: make sure slurm_init() is called before using
11   the Slurm API. This has been the case since version 20.11 (bsc#1209216).
12   Add-call-to-slurm_init-this-makes-sure-the-config-options-are-set.patch
13 </TARGET>
```

Note

Upstream CHANGELOG missing on purpose, requests adds patch file



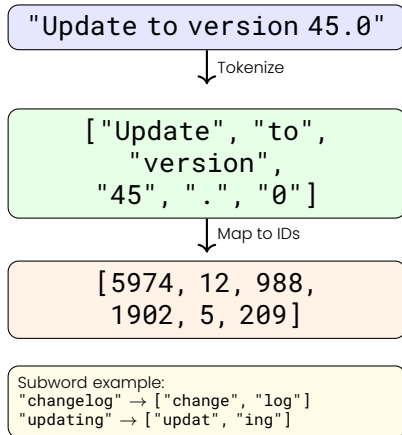
What is Tokenization?

Tokenization

Converts text into numerical representations that models can process.

How It Works

- Text split into subword units (tokens)
- Each token mapped to unique ID
- Vocabulary built from training corpus
- Unknown words broken into known subwords



Data Preprocessing & Tokenization

Special Token Handling

- Newline character (`\n`) added as special token
- Preserves structure in multi-line changelogs

Sequence Length Limits

- **Input:** Maximum 1024 tokens (spec diffs + changelog)
- **Target:** Maximum 512 tokens (generated changelog entry)
- Sequences exceeding limits are truncated

Padding Strategy

- Dynamic padding: sequences padded to **longest** in batch
- Padding tokens in labels replaced with `-100`
- Loss function ignores `-100` (no gradient from padding)



Fine-Tuning & Results

Untrained Models: Fine-Tuning is Necessary

Request: 1111873

small

- ```
1 Minor quality of life improvements seitens unserer contributors this release: comprised of the following:
 likewise: embodied and added translations: characterized by a re-written version of the gnome-music from
 44.0 to 45.0: utilised changelog: assemble changelog for package gnomes-music between 44.0 and 45.0 :
 räte changelog - rätze changelog. rätsche -Source0: https://download.gnome.org/source
```

**base**

- ```
1 :===== .tar.xz .tr.xs äcprès +Source0: https://download.gnome.org/sources/gnome-music/Retrouvez +=:
   Marinus Schraal ==Hungarian==rätsel près +Retrouvez +Source1: httpswww.gno
```

large

- ```
1 -music from 44.0 to 45.0: ===== erweisen erweisend erweisendlich Peter Deutscherweise Germanerweise
 Germangegenseitig erweiselich Germanerweise Englishgegenseitigerweise Englisherweiseerweise
 Englishgebührenpflichtig Germanerweise Polisherweise Polisherweise -Source0: --music ==-music-music
 erweise -erweise == =
```



# Untrained Models: Fine-Tuning is Necessary

Request: 1071207

**small**

```
1 -pdsh 2.34::-call-to-slurm_init-this-makes-sure-the-config-options-are-set.patch'] aeoe o ai epeadi в ил и
 пораии ирадиии орамителноо
```

**base**

```
1 structured changelog for package pdsh 2.34:increasing structured changelogging for package create structured
 change log for package build pcsh pkg 2.34.34:Auffinden Sie neue files: ['Add-call-to-slurm_init-this-
 makes-sure-the-config-options-are-set.patch']Währenden +%patch9 -p1: Add-call/slurstm_install-this
```

**large**

```
1 -p1
```



# Fine-Tuning Hyperparameters (Small Dataset)

## Common Parameters

- **Optimizer:** AdaFactor
- **Learning rate scheduler:** Cosine
- **Evaluation Split:** 10%
- **Dataset:** ~3,200 samples

## NVIDIA L40 GPU

- **Architecture:** Ada Lovelace
- **VRAM:** 48 GB GDDR6 (ECC)
- **CUDA Cores:** 18,176
- **TDP:** 300W

## Model-Specific Parameters

| Parameter  | t5-small | t5-base | t5-large |
|------------|----------|---------|----------|
| Batch Size | 30       | 10      | 3        |
| Max Steps  | 21,000   | 64,000  | 12,468   |
| Max Epochs | 200      | 200     | 12       |

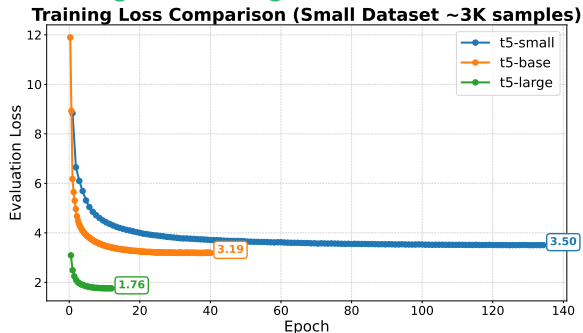
## Note

Batch sizes scaled inversely with model size due to memory constraints



# Training Results and Convergence (Small Dataset)

## Training Loss Progression



## Training Summary

| Model    | Steps  | Runtime  |
|----------|--------|----------|
| t5-small | 14,100 | 13h 08m  |
| t5-base  | 12,800 | 14h 23m  |
| t5-large | 12,468 | 308h 56m |

## Stopping Reason

- **t5-small/base**: Early stopping (patience=5)
- **t5-large**: Max steps reached



# Fine-Tuned Model: First Results

## Request 1111873 (gnome-music)

### t5-small:

- 1 - Update to version 45.0:
- 2 + Minor quality of life improvements.
- 3 + Updated translations.

### t5-base:

- 1 - Update to version 45.0:
- 2 + Minor quality of life improvements.

### t5-large:

- 1 - Update to version 45.0:
- 2 + Minor quality of life improvements.

## Request 1071207 (pdsh patch)

### t5-small:

- 1 - Add add-call-to-slurm\_init-this-makes-
- 2 sure-the-config-options-are-set.patch

### t5-base:

- 1 - Add add-call-to-slurm\_init-this-makes-
- 2 sure-the-config-options-are-set.patch
- 3 \* Fixes build with newer version of SLE15

### t5-large:

- 1 - Add-call-to-slurm\_init-this-makes-sure-
- 2 the-config-options-are-set.patch



# Fine-Tuned Models: Results Discussion

## Good

- No unrelated gibberish
- Mostly reflects the changes

## Ugly

### - Overfitting:

```
1 Updated translations
```

### - Hallucinations:

```
1 Fixes build with newer version of SLE15
```

- Other samples show similar errors or are worse
- Bigger dataset needed

## Overfitting

Small model summarizes a list of languages to "Updated translations"?

## Hallucination

Input doesn't contain any SLE15:

```
1 create structured changelog for package pdsh 2.34:
2 new files: ['Add-call-to-slurm_init-this-makes-sure-the
 -config-options-are-set.patch']
3 changes in spec file:
4 +Patch9: Add-call-to-slurm_init-this-makes-sure-the-
 config-options-are-set.patch
5 +%patch9 -p1
```



# Fine-Tuned Models: Scaling Up the Data

## Problem

- Collection of dataset took two days
- Approximately 1 in 10 requests go to `openSUSE:Factory` and are accepted
- `osc request list -P openSUSE:Factory -s accepted` times out

## Solution

- Adrian Schröter provided a list of **all** accepted requests
- Could easily create a list of 30,000 samples





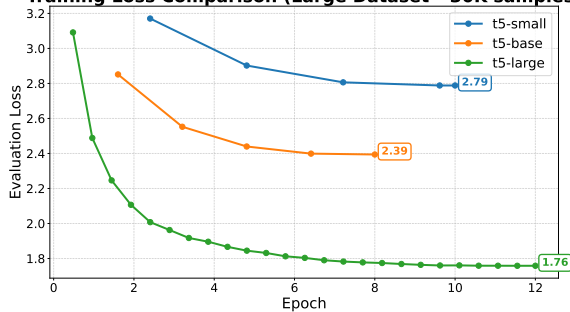
# Training Results and Convergence (Large Dataset)

## Training Summary

| Model    | Steps  | Runtime |
|----------|--------|---------|
| t5-small | 2,080  | 1h 52m  |
| t5-base  | 2,496  | 4h 56m  |
| t5-large | 12,468 | 23h 56m |

## Training Loss Progression

Training Loss Comparison (Large Dataset ~30K samples)



## Stopping Reason

- **t5-small**: Reached max epochs (10)
- **t5-base**: Reached max epochs (8)
- **t5-large**: Max steps reached (12)



# Fine-Tuned Models: Large Dataset Results

## Request 1111873 (gnome-music)

### t5-small:

```
1 - Update to version 45.0:
2 * Minor quality of life improvements
3 * Thanks to our contributors this release
```

### t5-base:

```
1 - Update to version 45.0:
2 + Minor quality of life improvements.
```

### t5-large:

```
1 - Update to version 45.0:
2 + Minor quality of life improvements.
```

## Request 1071207 (pdsh patch)

### t5-small:

```
1 - Add-call-to-slurm_init-this-makes-sure-
2 the-config-options-are-set.patch
```

### t5-base:

```
1 - Add Add-call-to-slurm_init-this-makes
2 sure the configuration options are
3 set.patch
```

### t5-large:

```
1 - Add-call-to-slurm_init-this-makes-sure-
2 the-config-options-are-set.patch
```



# Fine-Tuned Models: Large Dataset Discussion

## Good

- No hallucinations
- Small model includes too much text as it lacks semantic understanding

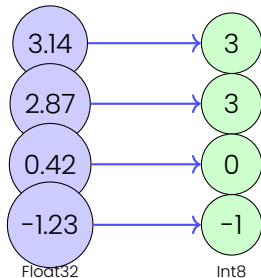
## Bad

**Base** and **large** models are too large to run on laptops (> 1 GB)

## Solution

Use quantization to reduce model size and runtime

### Quantization



# Quantization & Deployment

# What is Quantization?

## Quantization

Reduces numerical precision of weights from high precision (Float32) to lower (Int8).

## Why It Works

- Neural networks are **over-parametrized**
- Robust features tolerate small numerical errors
- Weight values cluster
- Activations follow predictable distributions

## Benefits

- **Smaller size:** Float32  $\rightarrow$  Int8 = 4 $\times$  reduction
- **Faster inference:** Integer operations are faster
- **Lower memory:** Fits on consumer hardware
- **Minimal accuracy loss:** Usually <2% degradation



# Quantization Implementation

## CTranslate2 vs. Ollama

- **Ollama**: Optimized for decoder-only models (GPT, LLaMA)
- **CTranslate2**: Optimized for encoder-decoder models (T5, BART)
- T5's bidirectional encoder requires specialized attention mechanisms

## INT8 Quantization

- Float32 (4 bytes) → Int8 (1 byte)
- **Result**: ~4× smaller model size
- **t5-base**: 892 MB → ~220 MB
- **t5-large**: 2.75 GB → ~690 MB

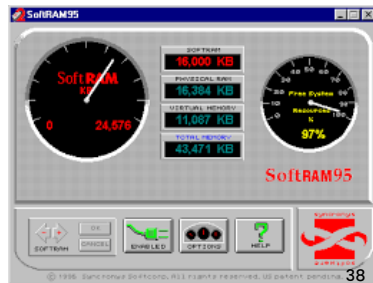


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## Compression Results

| Model    | Size   |
|----------|--------|
| t5-small | 60 MB  |
| t5-base  | 220 MB |
| t5-large | 690 MB |

All models now fit in typical laptop RAM!



# Wrap-up

## Model

- Model and quantized versions uploaded to Hugging Face
- <https://huggingface.co/mslacken/t5-finetune-changelog>
- FP32 model (full precision) for ctranslate2

## Integration

- Simple Python integration available on GitHub
- <https://github.com/openSUSE/changelog-gen>
- [obs://science:machinelearning/python-changelog-gen](https://obs://science:machinelearning/python-changelog-gen)



# Future Plans – LabsConf 2027

## Initial Python Package Generation

- Many python packages available
- py2pack struggles with multiflavor python builds
- More python packages sensible?

## Factory acceptance bot

- Check if package is fit for factory
- Package conflicts can't be caught
- Simpler answer, yay or nay



## Cross distribution packaging

- Would be immensely useful, if working
- Too complicated to get it working?
- Works both ways!





Thank you for your attention!



# Used Software & Availability

## Core Training Packages

- `python-torch`
- `python-transformers` - Hugging Face T5
- `python-tokenizers` - Fast tokenization
- `python-safetensors` - Model serialization

## Inference & Deployment

- `ctranslate2` - Quantization & fast inference

## openSUSE:Factory Status

| Package                          | Status |
|----------------------------------|--------|
| <code>python-torch</code>        | ✓      |
| <code>python-transformers</code> | ✓      |
| <code>python-tokenizers</code>   | ✓      |
| <code>python-safetensors</code>  | ✓      |
| <code>ctranslate2</code>         | ✓      |

`python-changelog-gen:`  
`science:machinelearning`

### Note

All essential packages available in Factory but without acceleration!



# Disk Space Requirements

Total Space Used: **37 GB**

| Component                     | Size    |
|-------------------------------|---------|
| Virtual Environment (PyTorch) | 5.7 GB  |
| Training Code & Scripts       | <100 MB |
| Datasets (small + large)      | 137 MB  |
| <i>Fine-tuned Models:</i>     |         |
| Small dataset runs            | 13 GB   |
| Large dataset runs            | 13 GB   |
| Current workspace             | 5.6 GB  |

